

# Assessment Book — Newton's Laws of Motion

Physics | Grade 9-10

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## Questions

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Total marks                      4

Q1. (1 mark) A bus brakes suddenly and passengers lurch forward. Why?

- A. A forward force acts on them
- B. They continue moving while the bus slows
- C. Gravity increases during braking
- D. Air pressure pushes them forward

Q2. (3 marks) A 4 kg trolley experiences a net force of 12 N. Find its acceleration.

Working:

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## Answer Key

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Q1.

Correct: B. They continue moving while the bus slows

A. Targets the belief that motion requires a force.

C. Confuses weight with motion.

Q2.

Answer:  $3 \text{ m/s}^2$

Working:  $a = F/m = 12 \text{ N} / 4 \text{ kg} = 3 \text{ m/s}^2$

Rubric — Correct method, substitution, and units

Complete (3 marks): Correct value with units.

Partial (2 marks): Correct method, wrong units.

Minimal (1 marks): Formula stated only.